

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Led design of a multi-agent system using OpenAI Agents SDK, MCP function/tool-calling, and LLM-based tool use, giving operators a natural-language interface into internal workflows handling millions of daily transactions.
- Owned full-stack delivery of internal operational tooling end-to-end: REST APIs, database modeling, and a React/TypeScript front end, with a deployment-gated quality pipeline enforcing automated checks before every release.
- Architected AWS serverless infrastructure (Lambda, S3, DynamoDB, API Gateway) exposing production APIs at sub-200ms latency, supporting high-throughput distribution at scale.

QureAI

Jan 2024 – May 2025

ML Infrastructure Engineer

- Drove a RAG retrieval redesign improving recall 61% to 89% on a 2K-query eval set via hybrid dense-sparse retrieval and re-ranking.
- Led migration of synchronous model serving to async vLLM on GCP Cloud Run with autoscaling, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Architected and operated 25+ production Java Spring Boot microservices exposing REST APIs for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS) in a distributed, fault-tolerant architecture.
- Drove a redesign of a nightly batch data pipeline aggregating financial transactions, cutting downstream-blocking processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams, reducing production escalations 60% in one quarter through code reviews and design write-ups.

Projects

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus key-value store with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production eval platform with rubric-based scoring, drift detection, and per-call latency tracking across model versions; FastAPI backend, React/TypeScript dashboard, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Java, TypeScript, JavaScript, Python, SQL

Full-Stack: Java (Spring Boot), React, HTML/CSS, REST APIs, Microservices, Database Modeling (PostgreSQL, DynamoDB, NoSQL)

Agentic AI: MCP Function/Tool-Calling, OpenAI Agents SDK, LLM Tool Use, RAG

Tools & Cloud: Git, Docker, Kubernetes, CI/CD, AWS (Lambda, EC2, S3, RDS, SQS), GCP (Cloud Run)

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer